PATIENT ENCOUNTER COST AND RISK ANALYSIS

Healthcare Financial Risk Analysis

Where costs go uncovered, which patients drive spend, and what to do next.

SQL Server analysis and Power BI dashboard · Findings and recommendations

Objective and scope

OBJECTIVE

Evaluate financial risk across healthcare encounters.

Find the costs payers do not cover, the patients who drive spend, and the diagnoses and procedures behind both.

KEY QUESTIONS

- How much cost is left uncovered, and by which payers?
- Which patients have frequent high-cost encounters?
- Which diagnoses and procedures carry the most cost?
- Where do encounters run long or repeat?

Source: Healthcare_Financial_risk database and Power BI dashboard

Data and method

01

Clean

Fix date and cost types, then check nulls, duplicates, negatives and invalid dates.

02

Measure

Uncovered cost is total claim cost minus payer coverage.

03

Segment

Cut by payer, reason code, demographics, procedure and organization.

04

Visualize

Publish the results in a Power BI dashboard for ongoing monitoring.

Tables: Encounters, Patients, Procedures, Payers, Organizations

Headline metrics

[$__]

Total uncovered cost

Claim cost minus payer coverage, all encounters.

[__%]

Cost from the top 10% of patients

The high-cost patient card on the dashboard.

[$__]

Average claim cost

Shown by encounter class, with minimum and maximum.

Fill the bracketed values from the Power BI dashboard cards

Uncovered cost by payer and reason

BY PAYER

Compare covered and uncovered cost for each payer.

Top payer by uncovered cost: [payer name], [$__]

BY REASON

Rank diagnoses by the cost left uncovered.

Top reason by uncovered cost: [reason], [$__]

Dashboard: Covered vs Uncovered Cost by Payer; Uncovered Cost by Reason Description

A few patients drive a large share of cost

DEFINITION

More than 3 encounters over $10,000 in one year.

Patients meeting this rule: [__], with total cost [$__].

PROFILE

Check gender, race, ethnicity and top reason codes.

Most common reason among them: [reason].

SQL sections 7 and 8; dashboard: High-Cost Patient Identification

Procedures, duration and geography

Cost trend

Average procedure cost over time and by diagnosis. Trend: [rising / flat / falling].

Long stays

Encounters over 24 hours by organization and class. Count: [__].

Repeat care

Patients with the same reason across several encounters. Count: [__].

Geography

Encounter volume and average cost by city. Busiest city: [city].

SQL sections 9 to 11; dashboard: Procedure Cost Trend, Patient Traffic by City

Recommendations

01 · Payers

Close the coverage gap

Review contracts with the payers that leave the most cost uncovered. Check claim denials and billing codes.

02 · Patients

Manage high-cost patients

Start care management for patients with more than 3 high-cost encounters a year, to prevent repeat admissions.

03 · Diagnoses

Target top cost drivers

Build standard care pathways for the few reason codes that account for most uncovered cost.

04 · Operations

Audit cost and duration

Investigate encounters over 24 hours, repeat procedures and large cost differences across organizations.

Each action maps to a view in the dashboard so progress can be tracked

Next steps

First 30 days

Validate the figures and name an owner for each recommendation.

Days 30 to 60

Open payer reviews and start care management for the high-cost group.

Days 60 to 90

Measure change in uncovered cost on the dashboard and refresh it monthly.

Questions and discussion